



Social Media Engagement Analysis Project

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Why This Project Is Very Important

Solves a Real Business Problem

Most companies spend huge budgets on social media, but struggle to answer:

- Which platform actually performs best?
- What type of content drives engagement?
- How do emotions influence user behavior?

This project directly answers these questions using data, not assumptions.

👉 Impact: Helps businesses optimize marketing spend and content strategy.



Introduction

- I analyzed 12,000+ social media posts using Python, SQL, and Power BI to uncover how platforms, sentiments, and emotions shape user engagement.
- The insights challenged common assumptions and revealed where brands truly win attention.



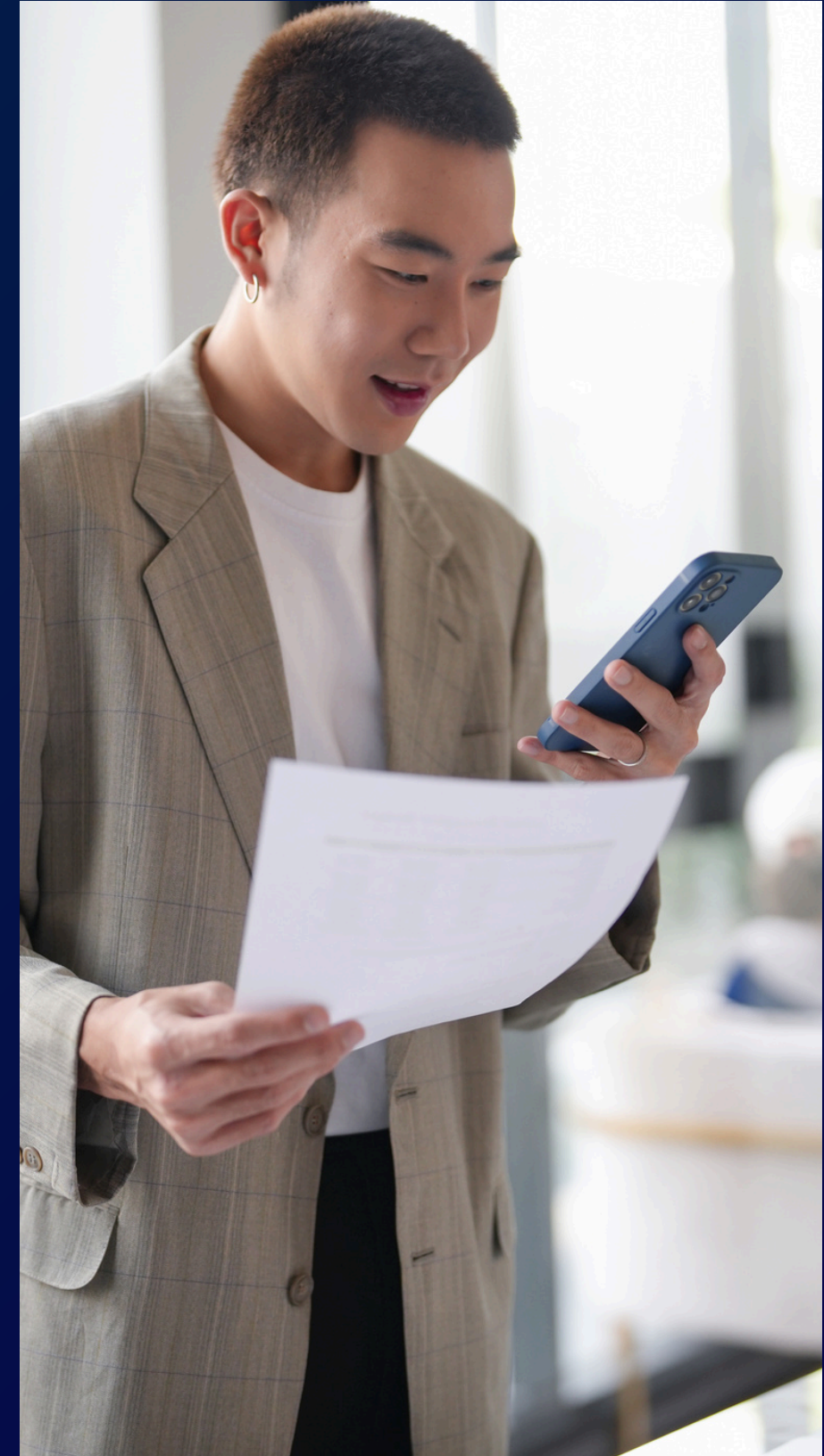
Platform-wise Engagement Performance

- Instagram and Facebook lead engagement, both achieving the highest average engagement rate (~0.29).
 - YouTube and Reddit follow closely (~0.28), indicating consistent audience interaction.
 - Twitter records the lowest engagement (~0.26), suggesting weaker content resonance or faster content decay.
- 📌 **Insight:** Visual-heavy and community-driven platforms outperform text-heavy platforms in engagement.



Engagement Rate Distribution

- The engagement rate is clustered within a narrow range, indicating stable but competitive engagement levels.
 - Very few extreme high-engagement posts exist, implying incremental optimization opportunities rather than viral dependency.
- 📌 **Insight: Consistent posting strategy works better than relying on viral content.**





Sentiment vs Engagement Relationship

- Negative sentiment surprisingly drives the highest engagement (40.74%).
- Positive sentiment follows closely (38.03%), while neutral content lags significantly (21.22%).

📌 **Insight:** Emotionally charged content (positive or negative) captures attention far more effectively than neutral messaging.



Emotion Type Distribution

- Posts are almost evenly distributed across emotions: Sad, Excited, Confused, Happy, and Angry.
 - No single emotion dominates volume, indicating diverse emotional storytelling.
-  **Insight: Balanced emotional content prevents audience fatigue and maintains sustained interest.**



Overall Social Media Health Snapshot

- **Total Posts: ~12,000 (strong dataset reliability)**
 - **Average Engagement Rate: ~27.8%**
 - **Positive Sentiment Percentage: ~40%**
- 📌 Insight: The brand or content ecosystem maintains healthy engagement with strong emotional influence.**

Q1: Which social media platforms generate higher engagement?

```
select platform,  
ROUND(AVG(engagement_rate)::NUMERIC,2) as  
avg_engagement_rate  
FROM socialusers  
GROUP BY platform  
ORDER BY avg_engagement_rate DESC;
```

	platform 	avg_engagement_rate 
	text	numeric
1	instagram	0.29
2	facebook	0.29
3	youtube	0.28
4	reddit	0.28
5	twitter	0.26

Q2: Which campaigns drive maximum user interaction?

```
select campaign_name,  
SUM(likes_count+shares_count+comments_count) as total_engagement  
FROM socialusers  
GROUP BY campaign_name  
ORDER BY total_engagement DESC  
limit 5;
```

	campaign_name  text	total_engagement  numeric
1	CyberMonday	2291420
2	LocalTouchpoints	2286975
3	InnovationX	2229252
4	LoyaltyRewards	2211268
5	BackToSchool	2203893

Q3: Which days of the week are best for posting content?

```
select day_of_week,  
ROUND(AVG(likes_count+shares_count+commen  
ts_count)::NUMERIC,2) AS avg_total_engagement  
FROM socialusers  
GROUP BY day_of_week  
ORDER BY avg_total_engagement DESC;
```

	day_of_week text	avg_total_engagement numeric
1	Sunday	4017.27
2	Saturday	4015.57
3	Wednesday	4015.49
4	Tuesday	4014.41
5	Thursday	3998.52
6	Monday	3994.57

Q4: Detect ineffective content that reaches users but doesn't engage them.

```
select post_id, impressions, engagement_rate
FROM socialusers
WHERE impressions > (select AVG(impressions)
FROM socialusers) AND
engagement_rate < (select
AVG(engagement_rate) FROM socialusers);
```

	post_id text	impressions bigint	engagement_rate double precision
1	vkmervg4ioos	52764	0.05086
2	wu063ci6mi...	57406	0.04442
3	9q2i3owe4tv9	85296	0.03274
4	gkyad7r8w6...	61572	0.06759
5	86ebsz8kvs...	98060	0.04844
6	w00na2lkf05v	52167	0.04205

Q5: Business Use Case: Understand brand perception across social media.

```
select brand_name,  
ROUND(AVG(user_past_sentiment_avg)::NUMERIC, 3) AS avg_sentiment_score,  
ROUND(AVG(likes_count+shares_count+comments_count)::NUMERIC, 2) AS avg_engagement  
FROM socialusers  
GROUP BY brand_name  
ORDER BY avg_sentiment_score DESC;
```

	brand_name 	avg_sentiment_score 	avg_engagement 
	text	numeric	numeric
1	toyota	0.033	3978.77
2	amazon	0.015	4120.42
3	microsoft	0.006	3927.78
4	samsung	-0.001	4063.00
5	pepsi	-0.001	3995.43
6	nike	0.002	3051.62

Strategic Recommendations (Business & Marketing Ready)

1. Platform-Specific Content Strategy

- Instagram & Facebook:**
- Focus on visual storytelling, reels, and carousel posts.**
- Twitter:**
- Improve engagement using polls, trending hashtags, and short emotional hooks.**
- YouTube & Reddit:**
- Leverage long-form value content and discussion-based posts.**

Strategic Recommendations (Business & Marketing Ready)

2. Emotion-Driven Content Optimization

- Use controlled negative sentiment (problem statements, pain points, debates) to drive conversations.**
- Reinforce positive emotions with success stories, transformations, and inspirational narratives.**
- Reduce overuse of neutral informational content or combine it with emotional framing.**

Strategic Recommendations (Business & Marketing Ready)

3. Engagement Growth Strategy

- Identify top-performing sentiment + platform combinations and replicate patterns.**
- Run A/B testing on emotional tone to measure engagement lift.**
- Use emotion-based tagging to personalize future content recommendations.**

Strategic Recommendations (Business & Marketing Ready)

4. Community & Brand Trust Building

- Monitor negative sentiment carefully to avoid brand risk.**
- Respond actively to high-engagement negative posts to convert criticism into trust.**
- Encourage user-generated content driven by emotional storytelling.**

Strategic Recommendations (Business & Marketing Ready)

5. Data-Driven Decision Making

- Integrate sentiment analysis into campaign planning.**
- Prioritize emotion-aware KPIs over vanity metrics like views.**
- Automate dashboards for real-time engagement monitoring.**

Conclusion

This project successfully demonstrates how data-driven analysis combined with sentiment and emotion insights can transform social media performance. By analyzing 12,000+ posts across multiple platforms, the study revealed that emotionally charged content—both positive and negative—drives significantly higher engagement compared to neutral messaging.

Through the integrated use of Python for data cleaning, SQL for analysis, and Power BI for interactive visualization, the project delivers clear, actionable insights that enable businesses to optimize platform-specific strategies, improve audience engagement, and make informed marketing decisions. Overall, this analysis highlights the importance of emotion-aware, insight-led content strategies in today's competitive digital landscape.



Thank You

